

J2
USB_C_Receptacle_USB2.0

Note: while we can monitor the CC pins to check for available current, the battery charger will always try to draw whatever current it needs

The schematic shows the internal wiring of the USB-C receptacle J2. The yellow box represents the connector housing. Pins are labeled as follows:

- VBUS (A4) connects to USB_CC1.
- CC1 (A5) and CC2 (B5) connect through resistors R22 (5.1k) and R23 (5.1k) to GND.
- D- (A7) and D+ (B7) connect to USB_D- and USB_D+ respectively.
- SBU1 (A8) and SBU2 (B8) are crossed out with blue X's.

External components and connections include:

- A green wire labeled S1 connects the SHIELD pin to a terminal block.
- A red wire labeled A1 connects the GND pin to another terminal block.
- A resistor R19 (DNP) is connected between the two terminal blocks.
- The bottom terminal block is connected to a common GND point.

Mux supposedly has internal protection diodes on Y+/-

+3.3V

10k R24

USB_MUX_SEL

USB_D+ USB_D-

1 2 3 4 5 6 7 8 9 10

D+ D- S HSD1- HSD1+ HSD2- HSD2+ OE VCC

U7 NLA57242MUTBG

PI_USB_MUXED_D+ PI_USB_MUXED_D- SMC_USB_MUXED_D+ SMC_USB_MUXED_D-

+3.3V

C23 100n

GND

Id: 1/1